



Intel Lunar Lake Chip

Powering new age of AI PCs at scale

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At the Intel Technology Tour in Taiwan, the air buzzed with anticipation as Yaron Alankry, vice president of Client Computing Group, and Arik Gihon, senior principal engineer of SoC Architecture, unveiled the architecture behind Lunar Lake processors. With promises of unparalleled performance and power efficiency, Lunar Lake is set to redefine the landscape of next-generation client products for Intel-powered

devices later this year. Intel's Lunar Lake architecture is designed to deliver breakthrough x86 power efficiency, a massive leap in graphics capabilities for an exceptional mobile gaming experience, never-before-seen AI compute on Intel devices, and exceptional hybrid core performance. According to the chip manufacturer, Lunar Lake isn't just an annual refresh to last year's Meteor Lake, but an ambitious project underscoring Intel's commitment to pushing

the boundaries of what's possible in chip design and performance.

Lunar Lake Overview

The heart of Lunar Lake's innovation lies in its microarchitecture, which consists of two unique tiles connected through Intel's industry-leading Foveros packaging technology, along with memory on package. The Compute Tile houses the latest-generation Efficient-cores (E-cores) and Performance-cores (P-cores), both introducing new microarchitecture enhancements for unprecedented x86 efficiency. It also includes the new Xe2 GPU, NPU 4, and IPU.

Compute Tile

- **Xe2 GPU:** Boasting over 80% more gaming performance and more than five times the AI throughput compared to the previous generation, the Xe2 GPU delivers up to 67 tera-operations per second (TOPS). Its new microarchitecture for display and media engines ensures a significantly improved visual experience compared to the older-gen Xe GPU.